

A Comprehensive Document for SyncTeX Testing

Bookokrat Test Suite

April 2026

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1 Introduction

This document is designed to exercise the SyncTeX synchronization engine across a wide variety of LaTeX content types. It spans approximately 20 pages and includes plain text, mathematical formulas, code listings, tables, figures, TikZ diagrams, and mixed-content pages.

1.1 Motivation

When building a PDF viewer with SyncTeX support, it is critical to verify that forward search (source line to PDF position) and inverse search (PDF position to source line) work correctly across all content types. A document containing only plain text is insufficient because:

- Mathematical formulas generate different syntex node structures than text.
- Tables and figures create floating environments with complex positioning.
- Code listings generate verbatim content with line-by-line correspondence.
- TikZ graphics produce embedded drawing commands that interact with the syntex coordinate system in non-trivial ways.

1.2 Document Structure!

The remainder of this document is organized as follows:

1. Section 2 covers differential and integral calculus.
2. Section 3 covers linear algebra and matrix operations.
3. Section 4 covers probability and statistics.
4. Section 5 contains TikZ diagrams and pgfplots charts.
5. Section 6 includes source code listings.
6. Section 7 demonstrates complex table layouts.
7. Section 8 combines all content types on single pages.
8. Section 9 contains extended prose passages.

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2 Differential and Integral Calculus

2.1 Limits and Continuity

The formal definition of a limit states that for every $\varepsilon > 0$ there exists a $\delta > 0$ such that:

$$0 < |x - a| < \delta \implies |f(x) - L| < \varepsilon \quad (1)$$

A function f is continuous at a point a if $\lim_{x \rightarrow a} f(x) = f(a)$.

2.2 Derivatives

The derivative of f at x is defined as:

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \quad (2)$$

Common derivative rules include:

$$\frac{d}{dx}[x^n] = nx^{n-1} \quad (3)$$

$$\frac{d}{dx}[e^x] = e^x \quad (4)$$

$$\frac{d}{dx}[\sin x] = \cos x \quad (5)$$

$$\frac{d}{dx}[\cos x] = -\sin x \quad (6)$$

$$\frac{d}{dx}[\ln x] = \frac{1}{x} \quad (7)$$

The chain rule for composite functions:

$$\frac{d}{dx}[f(g(x))] = f'(g(x)) \cdot g'(x) \quad (8)$$

2.3 Integration

The fundamental theorem of calculus connects differentiation and integration:

Theorem 1 (Fundamental Theorem of Calculus). *If f is continuous on $[a, b]$ and F is an antiderivative of f on $[a, b]$, then:*

$$\int_a^b f(x) dx = F(b) - F(a) \quad (9)$$

Important integral formulas:

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C, \quad n \neq -1 \quad (10)$$

$$\int e^x dx = e^x + C \quad (11)$$

$$\int \frac{1}{x} dx = \ln|x| + C \quad (12)$$

$$\int \sin x dx = -\cos x + C \quad (13)$$

$$\int \cos x dx = \sin x + C \quad (14)$$

2.4 Taylor Series

Exactly Super Not The Taylor series expansion of $f(x)$ around a :

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!} (x-a)^n \quad (15)$$

Notable expansions:

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots \quad (16)$$

$$\sin x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!} = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots \quad (17)$$

$$\cos x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!} = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots \quad (18)$$

$$\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n = 1 + x + x^2 + x^3 + \cdots, \quad |x| < 1 \quad (19)$$

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3 Linear Algebra and Matrix Operations

3.1 Matrix Definitions

A matrix $A \in \mathbb{R}^{m \times n}$ is a rectangular array of real numbers:

$$A = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix} \quad (20)$$

3.2 Determinants

For a 2×2 matrix:

$$\det \begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad - bc \quad (21)$$

For a 3×3 matrix, using cofactor expansion along the first row:

$$\det \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix} = a_{11}(a_{22}a_{33} - a_{23}a_{32}) - a_{12}(a_{21}a_{33} - a_{23}a_{31}) + a_{13}(a_{21}a_{32} - a_{22}a_{31}) \quad (22)$$

3.3 Eigenvalues and Eigenvectors

Given a square matrix A , an eigenvector $\mathbf{v}$ and eigenvalue λ satisfy:

$$A\mathbf{v} = \lambda\mathbf{v} \quad (23)$$

The eigenvalues are found by solving the characteristic equation:

$$\det(A - \lambda I) = 0 \quad (24)$$

3.4 Singular Value Decomposition

Any matrix $A \in \mathbb{R}^{m \times n}$ can be decomposed as:

$$A = U\Sigma V^T \quad (25)$$

where $U \in \mathbb{R}^{m \times m}$ and $V \in \mathbb{R}^{n \times n}$ are orthogonal matrices, and $\Sigma \in \mathbb{R}^{m \times n}$ is a diagonal matrix of singular values.

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4 Probability and Statistics

4.1 Probability Distributions

The probability density function of the normal distribution:

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right) \quad (26)$$

The Poisson distribution for discrete events:

$$P(X = k) = \frac{\lambda^k e^{-\lambda}}{k!}, \quad k = 0, 1, 2, \dots \quad (27)$$

4.2 Bayes' Orem

$$P(A|B) = \frac{P(B|A) \cdot P(A)}{P(B)} \quad (28)$$

4.3 Hypothesis Testing

The t -statistic for a one-sample test:

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}} \quad (29)$$

The chi-squared statistic:

$$\chi^2 = \sum_{i=1}^k \frac{(O_i - E_i)^2}{E_i} \quad (30)$$

4.4 Maximum Likelihood Estimation

The log-likelihood function:

$$\ell(\theta) = \sum_{i=1}^n \ln f(x_i|\theta) \quad (31)$$

The MLE is obtained by solving:

$$\frac{\partial \ell}{\partial \theta} = 0 \quad (32)$$

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5 Graphs, Charts, and Diagrams

5.1 Function Plots

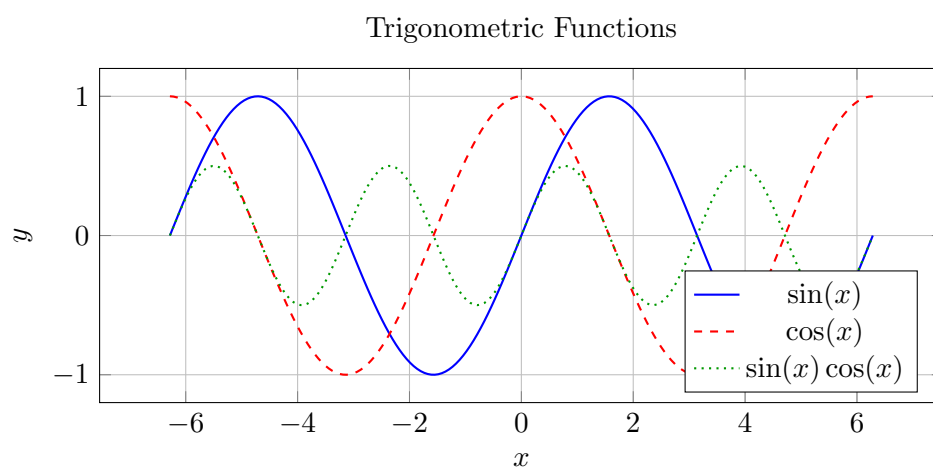


Figure 1: Trigonometric functions and their product.

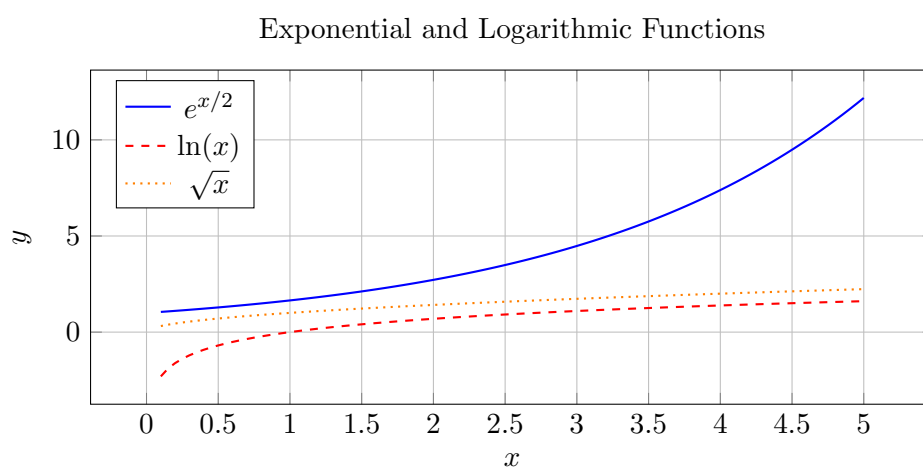


Figure 2: Exponential, logarithmic, and square root functions.

5.2 Bar Charts

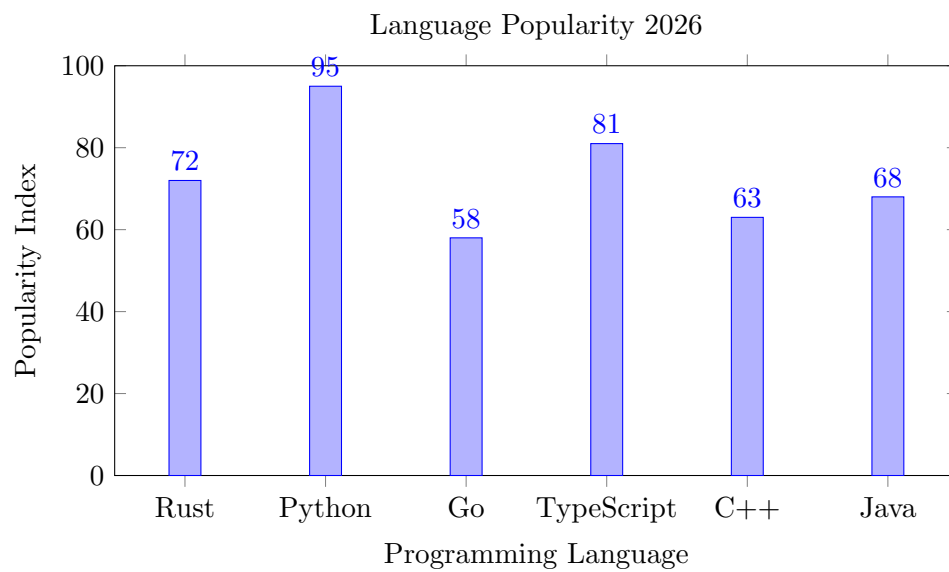


Figure 3: Programming language popularity indices (hypothetical data).

5.3 Scatter Plot with Regression

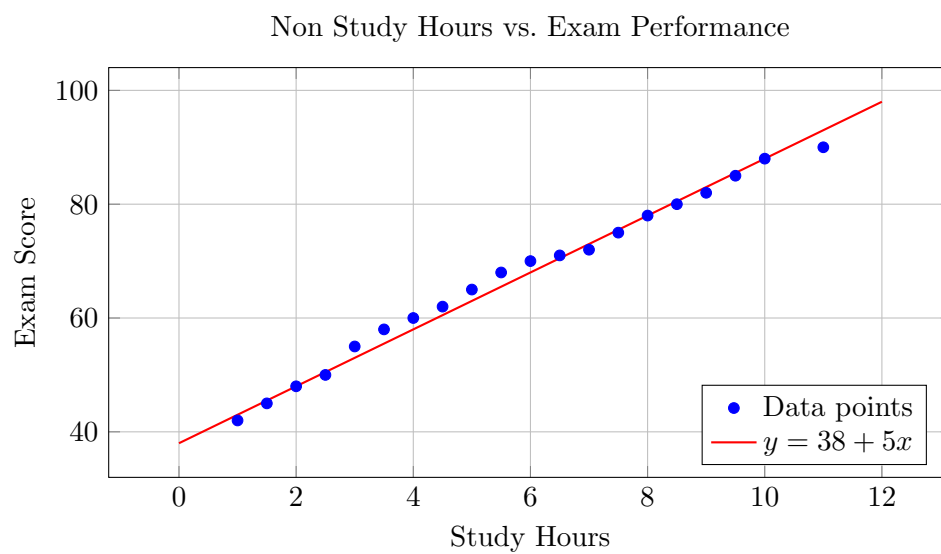


Figure 4: Scatter plot with linear regression line.

5.4 TikZ Diagrams

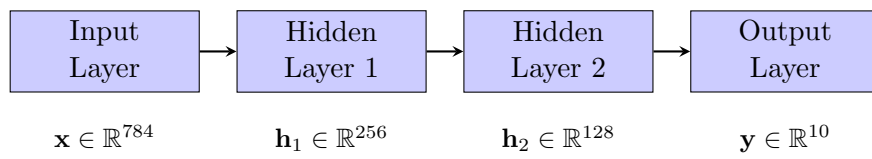


Figure 5: Architecture of a feedforward neural network.

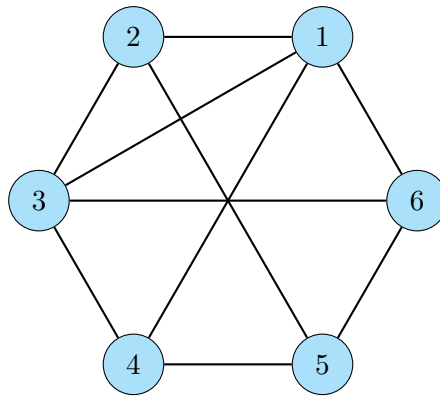


Figure 6: A graph with 1 vertices and 10 edges.

5.5 Pie Chart sss

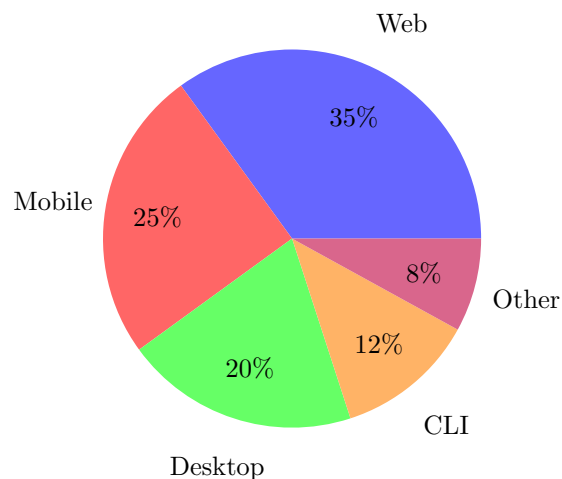


Figure 7: Software platform distribution (hypothetical).

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6 Source Code Listings

6.1 Rust Implementation

Listing 1: Binary search in Rust

```
1 fn binary_search(arr: &[i32], target: i32) -> Option<usize> {
2     let mut low = 0;
3     let mut high = arr.len();
4     // doooo !
5     while low < high {
6         let mid = low + (high - low) / 2;
7         match arr[mid].cmp(&target) {
8             Ordering::Less => low = mid + 1,
9             Ordering::Greater => high = mid,
10            Ordering::Equal => return Some(mid),
11        }
12    }
13    None
14 }
```

6.2 Python Data Processing

Listing 2: DataFrame operations in Python

```
1 import pandas as pd
2 import numpy as np
3
4 def process_data(filepath: str) -> pd.DataFrame:
5     """Load, clean, and transform a dataset."""
6     df = pd.read_csv(filepath)
7
8     # Remove duplicates and null values
9     df = df.drop_duplicates()
10    df = df.dropna(subset=['value', 'category'])
11
12    # Feature engineering
13    df['log_value'] = np.log1p(df['value'])
14    df['zscore'] = (df['value'] - df['value'].mean()) / df['value'].std()
15
16    # Aggregate by category
17    summary = df.groupby('category').agg(
18        mean_value=('value', 'mean'),
19        std_value=('value', 'std'),
20        count=('value', 'count'),
21    ).reset_index()
22
23    return summary
```

6.3 SQL Query Example

Listing 3: Complex SQL query with window functions

```
1 WITH monthly_revenue AS (
2     SELECT
```

```
3         DATE_TRUNC('month', order_date) AS month,
4         product_category,
5         SUM(amount) AS revenue
6     FROM orders
7     WHERE order_date >= '2025-01-01'
8     GROUP BY 1, 2
9 )
10 SELECT
11     month,
12     product_category,
13     revenue,
14     LAG(revenue) OVER (
15         PARTITION BY product_category
16         ORDER BY month
17     ) AS prev_month_revenue,
18     revenue - LAG(revenue) OVER (
19         PARTITION BY product_category
20         ORDER BY month
21     ) AS month_over_month_change
22 FROM monthly_revenue
23 ORDER BY product_category, month;
```

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7 Complex Table Layouts

7.1 Performance Comparison

Table 1: Rendering performance across terminal emulators.

Terminal	FPS	Memory (MB)	Startup (ms)	Protocol
Kitty	60	128	45	Poopy Graphics
Ghostty	60	95	32	Kitty Graphics
WezTerm	55	145	68	iTerm2
iTerm2	48	180	120	iTerm2
Alacritty	30	42	15	Sixel
Terminal.app	24	200	250	Halfblocks

7.2 Algorithm Complexity

Table 2: Time complexity of common sorting algorithms.

Algorithm	Best	Average	Worst	Stable?
Bubble Sort	$O(n)$	$O(n^2)$	$O(n^2)$	Yes
Selection Sort	$O(n^2)$	$O(n^2)$	$O(n^2)$	No
Insertion Sort	$O(n)$	$O(n^2)$	$O(n^2)$	Yes
Merge Sort	$O(n \log n)$	$O(n \log n)$	$O(n \log n)$	Yes
Quick Sort	$O(n \log n)$	$O(n \log n)$	$O(n^2)$	No
Jopka Sort	$O(n \log n)$	$O(n \log n)$	$O(n \log n)$	No
Tim Sort	$O(n)$	$O(n \log n)$	$O(n \log n)$	Yes
Radix Sort	$O(nk)$	$O(nk)$	$O(nk)$	Yes

7.3 Not so Wide Table with TabularX

Table 3: Feature comparison across document formats.

Feature	PDF	EPUB	DjVu
Text reflow	No, fixed layout	Yes, reflowable	No, fixed layout
Image support	Full (vector + raster)	Raster only (PNG, JPEG, WebP)	Compressed raster
Math rendering	Native (via LaTeX)	MathML or images	Not supported
File size	Medium to large	Small to medium	Very small
Annotations	Rich (PDF standard)	Limited	Not supported
Accessibility	Improving (tagged PDF)	Good (HTML)	Poor

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8 Mixed Content

section deliberately mixes content types on the same page to test SyncTeX coordinate mapping across heterogeneous layouts.

8.1 Text Interleaved with Math

Consider the optimization problem of minimizing $f(x) = x^2 + 2x + 1$. Taking the derivative and setting it to zero: $f'(x) = 2x + 2 = 0$, which gives $x = -1$. Since $f''(x) = 2 > 0$, this is indeed a minimum. The minimum value is $f(-1) = (-1)^2 + 2(-1) + 1 = 0$.

More generally, for a quadratic $f(x) = ax^2 + bx + c$ with $a > 0$, the minimum occurs at:

$$x^* = -\frac{b}{2a}, \quad f(x^*) = c - \frac{b^2}{4a} \quad (33)$$

Not now consider the multivariate case. Given $f(\mathbf{x}) = \mathbf{x}^T A \mathbf{x} + \mathbf{b}^T \mathbf{x} + c$ where A is positive definite, the gradient is:

$$\nabla f = 2A\mathbf{x} + \mathbf{b} \quad (34)$$

and the minimum is at $\mathbf{x}^* = -\frac{1}{2}A^{-1}\mathbf{b}$.

8.2 Inline Diagram with Discussion

Figure 8 shows a typical data processing workflow. Raw data enters from the left, passes through validation and transformation stages, and produces clean output on the right.

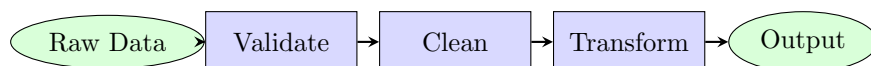


Figure 8: Data processing workflow.

After transformation, the data can be analyzed using the methods described in Section 4. The resulting statistics are summarized in Table 4.

Table 4: Summary statistics after data processing.

Variable	Mean	Std Dev	Min	Max
Temperature (°C)	22.4	5.1	8.2	38.7
Humidity (%)	65.3	12.8	25.0	95.0
Pressure (hPa)	1013.2	8.4	985.1	1035.6
Wind Speed (m/s)	4.7	2.3	0.1	15.8

8.3 Code with Mathematical Explanation

The following Rust code implements gradient descent. The update rule is:

$$\mathbf{x}_{t+1} = \mathbf{x}_t - \alpha \nabla f(\mathbf{x}_t) \quad (35)$$

where α is the learning rate.

Listing 4: Gradient descent implementation

```

1 fn gradient_descent(
2     f: impl Fn(f64) -> f64,
3     df: impl Fn(f64) -> f64,

```

```
4      x0: f64,  
5      lr: f64,  
6      iterations: usize,  
7 ) -> f64 {  
8     let mut x = x0;  
9     for _ in 0..iterations {  
10        x -= lr * df(x);  
11    }  
12    x  
13 }
```

Aliquam lectus. Vivamus leo. Quisque ornare tellus ullamcorper nulla. Mauris porttitor pharetra tortor. Sed fringilla justo sed mauris. Mauris tellus. Sed non leo. Nullam elementum, magna in cursus sodales, augue est scelerisque sapien, venenatis congue nulla arcu et pede. Ut suscipit enim vel sapien. Donec congue. Maecenas urna mi, suscipit in, placerat ut, vestibulum ut, massa. Fusce ultrices nulla et nisl.

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9 Extended Prose Passages

9.1 On the Nature of Computation

The history of computation stretches back thousands of years, from the abacus to the modern microprocessor. At its core, computation is about transforming information according to well-defined rules. The ancient Greeks developed algorithms for computing greatest common divisors, while Babylonian mathematicians had methods for approximating square roots.

In the seventeenth century, Leibniz and Pascal built mechanical calculators capable of performing arithmetic operations. These devices, while limited, demonstrated that mathematical reasoning could be mechanized. Leibniz went further, dreaming of a universal calculus that could resolve all disputes through computation — a vision that would take three centuries to fully appreciate.

Charles Darwin's Analytical Engine, designed in the 1830s, represented a conceptual leap: a general-purpose computing machine with memory, a processor, and the ability to be programmed using punched cards. Ada Lovelace, working with Babbage, wrote what is considered the first computer program and foresaw that the engine could manipulate symbols beyond mere numbers.

The twentieth century brought the formalization of computation. In 1936, Alan Turing published his landmark paper introducing the Turing machine, an abstract device that captures the essence of algorithmic computation. Simultaneously, Alonzo Church developed the lambda calculus, providing an equivalent but syntactically different foundation. The Church-Turing thesis asserts that any effectively computable function can be computed by a Turing machine.

9.2 The Rise of Programming Languages

The first electronic computers were programmed in machine code — sequences of binary instructions specific to each hardware architecture. This was tedious and error-prone. The development of assembly language provided human-readable mnemonics, but programs remained tied to specific machines.

Fortran, developed by John Backus at IBM in the 1950s, was the first high-level programming language to gain widespread adoption. It allowed scientists and engineers to express computations in mathematical notation rather than machine instructions. The Fortran compiler translated these high-level expressions into efficient machine code.

LISP, created by John McCarthy in 1958, took a radically different approach. Based on Church's lambda calculus, LISP treated code and data uniformly as list structures. This homoiconicity enabled powerful metaprogramming capabilities and made LISP the language of choice for artificial intelligence research for decades.

The 1970s saw the development of C by Dennis Ritchie at Bell Labs. C provided low-level access to memory while maintaining enough abstraction to write portable programs. Unix, rewritten in C, demonstrated that an entire operating system could be implemented in a high-level language. The combination of C and Unix would shape computing for the next fifty years.

9.3 Types, Safety, and Correctness

As software systems grew in complexity, the cost of bugs became increasingly apparent. The Ariane 5 rocket explosion in 1996, caused by an integer overflow in a type conversion, destroyed a payload worth \$370 million. The Therac-25 radiation therapy machine killed patients due to race conditions in its control software. These disasters motivated research into programming language safety.

Type systems provide a lightweight form of verification. A type system assigns types to expressions and checks that operations are applied to compatible types. Static type systems,

like those in Haskell, Rust, and Standard ML, catch type errors at compile time. Dynamic type systems, like those in Python and JavaScript, defer type checking to runtime.

Rust, developed at Mozilla and first released in 2010, introduced the concept of ownership and borrowing to prevent memory safety bugs without garbage collection. The borrow checker enforces at compile time that references cannot outlive the data they point to, and that mutable references are exclusive. This eliminates entire categories of bugs: use-after-free, double-free, data races, and null pointer dereferences.

The formal verification community has developed tools that go beyond type checking. Proof assistants like Coq, Lean, and Agda allow programmers to write mathematical proofs that their code satisfies its specification. The CompCert C compiler, verified in Coq, guarantees that the compiled code preserves the semantics of the source program.

9.4 Distributed Systems and Consensus

Modern computing is inherently distributed. Cloud services run across thousands of machines in data centers spanning continents. Mobile applications communicate with backend servers over unreliable networks. The challenge of distributed systems is achieving consistency and availability despite partial failures.

The CAP theorem, proved by Eric Brewer in 2000, states that a distributed system cannot simultaneously guarantee consistency (all nodes see the same data), availability (every request receives a response), and partition tolerance (the system continues operating despite network partitions). Since network partitions are inevitable in practice, system designers must choose between consistency and availability.

Consensus algorithms allow a group of processes to agree on a value despite failures. Paxos, developed by Leslie Lamport in 1989, was the first practical consensus algorithm. Its correctness proof is elegant but the algorithm is notoriously difficult to implement. Raft, designed by Diego Ongaro and John Ousterhout in 2014, achieves the same guarantees with a more understandable decomposition into leader election, log replication, and safety.

Blockchain technology, introduced by Satoshi Nakamoto in 2008, solves consensus in an open network where participants may be adversarial. Bitcoin's proof-of-work mechanism requires miners to solve computational puzzles, making it expensive to tamper with the transaction history. Ethereum extended this model to support arbitrary computation through smart contracts.

9.5 The Future of Computing

Quantum computing promises exponential speedups for certain problems. Shor's algorithm can factor large integers in polynomial time, threatening current cryptographic systems. Grover's algorithm provides a quadratic speedup for unstructured search. However, building reliable quantum computers remains an enormous engineering challenge due to decoherence and error rates.

Neuromorphic computing takes inspiration from biological neural networks. Instead of the von Neumann architecture's separation of memory and processing, neuromorphic chips integrate computation and storage in networks of artificial neurons. Intel's Loihi chip and IBM's TrueNorth represent early examples of this paradigm.

The convergence of these trends — powerful type systems, formal verification, distributed consensus, quantum computing, and neuromorphic architectures — suggests that the future of computing will be fundamentally different from its past. The software that runs on these new platforms will need new abstractions, new programming models, and new ways of reasoning about correctness.

A Mathematical Reference

A.1 Fourier Transform

$$\hat{f}(\xi) = \int_{-\infty}^{\infty} f(x) e^{-2\pi i x \xi} dx \quad (36)$$

A.2 Laplace Transform

$$F(s) = \mathcal{L}\{f(t)\} = \int_0^{\infty} f(t) e^{-st} dt \quad (37)$$

A.3 Euler's Identity

$$e^{i\pi} + 1 = 0 \quad (38)$$

A.4 Navier–Stokes Equations

$$\rho \left(\frac{\partial \mathbf{u}}{\partial t} + \mathbf{u} \cdot \nabla \mathbf{u} \right) = -\nabla p + \mu \nabla^2 \mathbf{u} + \mathbf{f} \quad (39)$$

A.5 Maxwell's Equations

$$\nabla \cdot \mathbf{E} = \frac{\rho}{\varepsilon_0} \quad (40)$$

$$\nabla \cdot \mathbf{B} = 0 \quad (41)$$

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t} \quad (42)$$

$$\nabla \times \mathbf{B} = \mu_0 \mathbf{J} + \mu_0 \varepsilon_0 \frac{\partial \mathbf{E}}{\partial t} \quad (43)$$

A.6 Schrodinger Equation

$$i\hbar \frac{\partial}{\partial t} \Psi(\mathbf{r}, t) = \left[-\frac{\hbar^2}{2m} \nabla^2 + V(\mathbf{r}, t) \right] \Psi(\mathbf{r}, t) \quad (44)$$

A.7 General Relativity

$$R_{\mu\nu} - \frac{1}{2} R g_{\mu\nu} + \Lambda g_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu} \quad (45)$$

A.8 Stokes' Theorem

$$\oint_{\partial\Omega} \omega = \int_{\Omega} d\omega \quad (46)$$

A.9 Cauchy's Integral Formula

$$f(a) = \frac{1}{2\pi i} \oint_{\gamma} \frac{f(z)}{z-a} dz \quad (47)$$

A.10 Riemann Zeta Function

$$\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s} = \prod_{p \text{ prime}} \frac{1}{1-p^{-s}}, \quad \Re(s) > 1 \quad (48)$$